



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

HORTICULTURE

PAPER 1 Multiple Choice

6024/1

1 hour

SPECIMEN PAPER

Additional materials:
Multiple Choice answer sheet,
Soft clean eraser,
Soft pencil (type B or HB is recommended),
Electronic calculator.

INSTRUCTIONS TO CANDIDATES

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet provided.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. Any rough working should be done in this booklet.

This specimen paper consists of 12 printed pages.

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1. A landscape design principle that creates proportionality among design elements is
 - A scale.
 - B unity.
 - C balance.
 - D focalization.

2. In old civilization, ornamental gardens were cultivated by
 - A royalty and the wealthy.
 - B landlords and peasants.
 - C peasants and commercial farmers.
 - D commercial farmers and small holder farmers.

3. Select the greatest factor that affects horticultural production.
 - A wind
 - B light
 - C rainfall
 - D temperature

4. Choose a socio-economic importance of horticulture.
 - A beautification
 - B nutritional value
 - C therapeutic value
 - D employment creation

5. Select horticultural equipment used in the processing of horticultural products.
- A Driers
 - B Planters
 - C Dusting machines
 - D Combine harvesters
6. The ability of a plant cell to regenerate the whole plant is termed as
- A germination.
 - B propagation.
 - C totipotency.
 - D layering.
7. Choose an advantage of tissue culture.
- A Production of exact copies of plants
 - B Unwanted traits will be produced
 - C Produce varied types of fruits
 - D Delayed maturation of plants
8. Identify a method commonly used in propagating oranges in Zimbabwe.
- A sucker
 - B cutting
 - C layering
 - D budding

9. The Kreb's cycle occurs in the
- A matrix.
 - B cristae.
 - C cytoplasm.
 - D mitochondria.
10. An apical meristem is formed from
- A lateral, intercalary and subapical.
 - B dermal, vascular and ground stem.
 - C calyptogens, periblemium and plerone.
 - D protoderm, procambium and ground stem.
11. How has climate change impacted small holder horticultural production in Zimbabwe?
- A increased yield per unit area
 - B reduced the return per dollar invested
 - C increased the return per dollar invested
 - D reduced demand for horticultural produce
12. Which irrigation system is used in the horticultural industry for frost protection?
- A drip irrigation
 - B flood irrigation
 - C trickle irrigation
 - D sprinkler irrigation

13. What is the effect of low soil pH on the following plant nutrients?

Ca, Mg concentration

Fe, Mn concentration

Increase

Decrease

Increase

Increase

Decrease

Increase

Decrease

Decrease

14. How does soil compaction affect the following properties?

Bulky density

Pore space

A increase

decrease

B decrease

decrease

C increase

increase

D decrease

increase

15. Air currents may accelerate the rate of transpiration by

A increasing humidity.

B raising the temperature.

C increasing CO₂ concentration.

D preventing water vapour accumulation.

16. The application rate for borox is **30g/10m²** calculate the quantity of fertilizer required in a greenhouse measuring **24 m x 50 m**.

A 0,25 kg

B 0, 36 kg

C 36, 0 kg

D 3, 6 kg

17. What is the reason for Zimbabwe banning certain pesticides from use?
- A low demand
 - B price fluctuations
 - C import challenges
 - D bio accumulations
18. Why is crop residue destruction important in horticulture?
- A breaks pest life cycle
 - B maintains field hygiene
 - C promotes nutrient recycling
 - D rests a field during off season
19. Transmission of horticultural plant diseases into the country is enhanced by
- A farm visits.
 - B high exports.
 - C high imports.
 - D retaining seeds.
20. Why is couch grass difficult to control?
- A It is deep rooted
 - B It has a trailing growth habit
 - C It produces many viable seeds
 - D It propagates by stolons and rhizomes

- 21.** Select a method commonly used by small holder horticultural farmers in the aeration of lawns.
- A** pruning
 - B** mowing
 - C** spiking
 - D** shaping
- 22.** Ornamental plant maintenance involves the following.
- A** planting and irrigation
 - B** pruning and irrigation
 - C** shaping and planting
 - D** pruning and harvesting
- 23.** Form as an element of landscaping refers to
- A** lines used in creating beds and walkways.
 - B** outdoor designs that balance size of buildings.
 - C** geometric shapes or combinations of shapes in landscaping.
 - D** varying textures that add to the atmosphere of an outdoor area.
- 24.** Interpret design principle of focalization.
- A** distribution of the design material
 - B** stability resulting from equal distribution of weight
 - C** presents an equal visual weight of elements to a viewer
 - D** creates interest and accent for a particular arrangement

25. The male organ of a flower is called
- A dioecious.
 - B gynaecium.
 - C androecium.
 - D monoecious.
26. What role do flowers play in offering condolences at funeral occasions?
- A they evoke happiness
 - B offer love and comfort
 - C offer comfort and support.
 - D they beautify the occasion.
27. Cut flowers are generally graded according to stem
- A size.
 - B length.
 - C diameter.
 - D base cut.
28. How is pot flower growth affected by the following temperature?

day temperature 25C

day temperature above 30C

- | | |
|--------------------------|------------------------|
| A low vegetative growth | low vegetative growth |
| B high vegetative growth | high vegetative growth |
| C high vegetative growth | stunted growth |
| D low vegetative growth | stunted growth |

29. Select a method of adding value to vegetables.

- A** Drying
- B** Thinning
- C** Harvesting
- D** Advertising

30. The following transactions were extracted from a vegetable farmer's diary. Use the information to answer question **30**.

15/03/23	Purchased fertilizers for	\$120-00
	Paid land preparation costs	\$ 130-00
20/03/23	Bought cabbage seedlings for	\$100-00
	Paid labour for transplanting	\$40-00
	Purchased pesticides for	\$50-00
5/05/15	Paid management costs	\$40-00
1/07/23	Paid harvesting labour costs	\$60-00
	Sold 3000 heads at \$0.50	\$1500-00

Calculate the total costs for the enterprise.

- A** 2040
- B** 1500
- C** 640
- D** 540

31. The reason for low vegetable consumption in Zimbabwe is
- A consumer preferences.
 - B unavailability of vegetables.
 - C low production levels of vegetables.
 - D high production cost of vegetables.
32. Select a method most commonly used by indigenous farmers in the preservation of vegetables.
- A boiling
 - B drying
 - C spicing
 - D frying
33. The main factor considered for apple production in the Eastern Highlands of Zimbabwe is
- A fertile soils in the region.
 - B ready market for apples.
 - C cool and humid conditions.
 - D favourable soil pH of 5.8 to 6.5.
34. What is the most important factor that determines mode of transport to the market for horticultural produce?
- A fuel cost
 - B road network
 - C product value
 - D product volume

35. The importance of fruit production to the country is
- A food.
 - B educational purposes.
 - C employment creation.
 - D foreign currency generation.
36. A propagation method suitable for commercial mango production is
- A suckering.
 - B budding.
 - C grafting.
 - D cutting.
37. When herbs on the landscape provide pleasing scents in the environment, they are classified as
- A culinary.
 - B ornament.
 - C fragrance.
 - D medicinal.
38. An appropriate production system for herb and spice in the green house is
- A raft water culture.
 - B the Gerick's system.
 - C ground culture and container culture.
 - D soilless culture and ground culture.

39. Which crop is propagated by cloves?

- A** onion
- B** garlic
- C** ginger
- D** mustard

40. Cat facing disorder in green pepper is caused by

- A** nutrient deficiency.
- B** both stamens and pistil.
- C** exposure of fruits to periods of drought before maturity.
- D** exposure of the plant to cool temperatures at the time of flower initiation.